ENVS-193DS_spring-2026_final

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GitHub Repository: ENVS-193DS Spring 2026 Final

Problem 1

a.

In Part 1, my co-worker used logistic regression. We can tell because the goal is to predict a probability (will an Avocet use this microhabitat or not, which is a yes/no outcome) from a continuous measurement (distance to the water edge in meters). Logistic regression is the right tool when your response variable is binary like this.

In Part 2, my co-worker used a chi-square test of independence. This is because both variables being compared are categorical — bird species (Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). A chi-square test tells us whether the two categorical variables are associated with each other or not.

b.

Part 1: American Avocets were more likely to be found in microhabitat sites that were closer to the water's edge. This suggests that how far a spot is from the water plays a role in whether or not an Avocet chooses to use it.

Part 2: The three bird species did not use the five habitat types equally. Each species tended to show up more in certain habitats than others, meaning the habitat a bird is found in is related to which species it is.

c.

One additional variable that could be worth looking at is water depth, measured in centimeters at each survey point using a ruler or depth probe. American Avocets are wading birds that forage by sweeping their bill through shallow water, so sites that are too deep or too dry may simply not be usable for feeding, which would directly affect whether or not Avocets choose to use a given microhabitat.

Problem 2

a.

```r
nests_clean <- nests |>
  select(case_control,
         height_cm,
         ant_species) |>
  filter(ant_species == "AB" |
         ant_species == "RRB" |
         ant_species == "BBR") |>
  mutate(ant_species = factor(ant_species,
                              levels = c("AB",
                                         "RRB",
                                         "BBR"))) |>
  mutate(full_name = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  ))

str(nests_clean)
```

```
'data.frame':	270 obs. of  4 variables:
 $ case_control: int  1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm   : num  392 310 513 154 324 ...
 $ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ full_name   : chr  "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae"
```

```r
slice_sample(nests_clean, n = 10)
```

```
   case_control height_cm ant_species               full_name
1             0     153.5         RRB   Crematogaster mimosae
2             0     117.5         RRB   Crematogaster mimosae
3             0     109.0         RRB   Crematogaster mimosae
4             0     199.5         BBR Crematogaster nigriceps
5             0     154.0         RRB   Crematogaster mimosae
6             0     193.5         RRB   Crematogaster mimosae
7             0     216.0         RRB   Crematogaster mimosae
8             0     178.0         RRB   Crematogaster mimosae
9             0     130.0         RRB   Crematogaster mimosae
10            0     131.0         RRB   Crematogaster mimosae
```

b.

The 1s and 0s in the case_control column represent whether a given tree contained a bird nest (1) or was a nearby available tree with no nest present (0). This is a binary outcome used to compare the characteristics of trees that birds actually chose to nest in versus trees that were available but unused.

c.

Tree height (height_cm, continuous variable measured in centimeters): As tree height increases, the probability of nest occurrence would likely increase because taller trees may offer more structural complexity and concealment, making them more attractive and safer nesting locations. The authors note that all nests found in the study were located in live trees more than 1.5 m in height, suggesting a minimum height threshold for nest use. (Methods, paragraph 2; Results, paragraph 1)

Acacia ant species (ant_species, categorical variable with 3 levels: C. sjostedti, C. mimosae, and C. nigriceps): The ant species occupying a tree could influence the probability of nest occurrence because different species vary in how aggressively they defend their host trees against predators. C. mimosae and C. nigriceps are aggressive defenders against both mammalian and insect herbivores, and birds nesting in those trees may benefit from protection against nest predators. In contrast, C. sjostedti provides only minimal protection, making those trees potentially less safe or attractive for nesting birds. Additionally, C. nigriceps prunes the apical buds of its host tree, creating a denser canopy that likely provides extra concealment from predators such as snakes, mesocarnivores, and raptors. (Introduction, paragraphs 2 and 4; Discussion, paragraphs 2 and 3)

d.

Model Number	Description
Model 1	Probability of nest occurrence predicted by tree height and acacia ant species, no interaction
Model 2	Probability of nest occurrence predicted by tree height and acacia ant species, with interaction between tree height and acacia ant species

e.

```r
# Model 1: no interaction
model1 <- glm(case_control ~ height_cm + ant_species,
              data = nests_clean,
              family = binomial)

# Model 2: with interaction
model2 <- glm(case_control ~ height_cm * ant_species,
              data = nests_clean,
              family = binomial)
```

f.

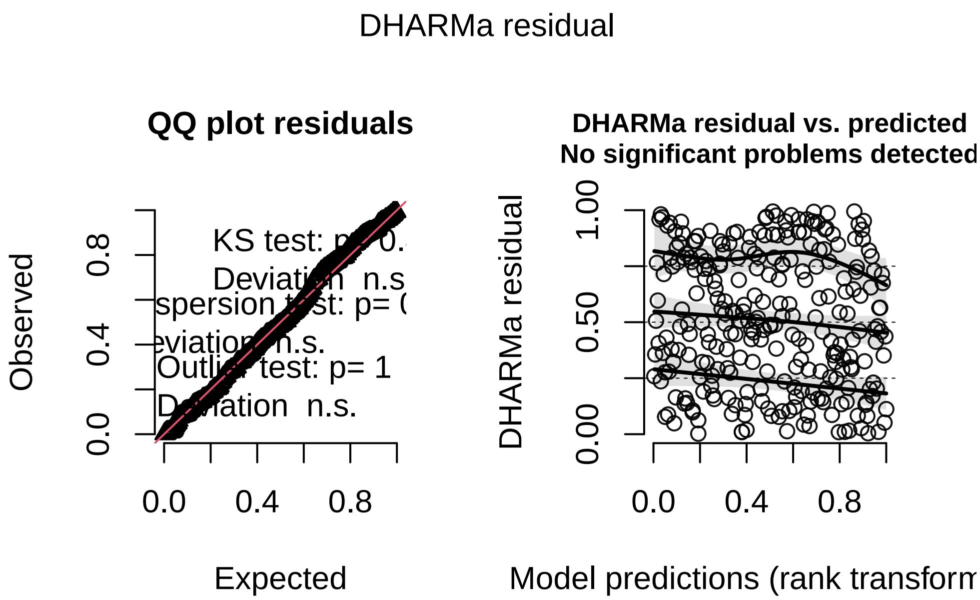
```r
AIC(model1, model2)
```

```
       df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) was the model predicting probability of bird nest occurrence from tree height and acacia ant species with an interaction term (AIC = 237.80), as it had a lower AIC value than the model without an interaction term (AIC = 258.74).

g.

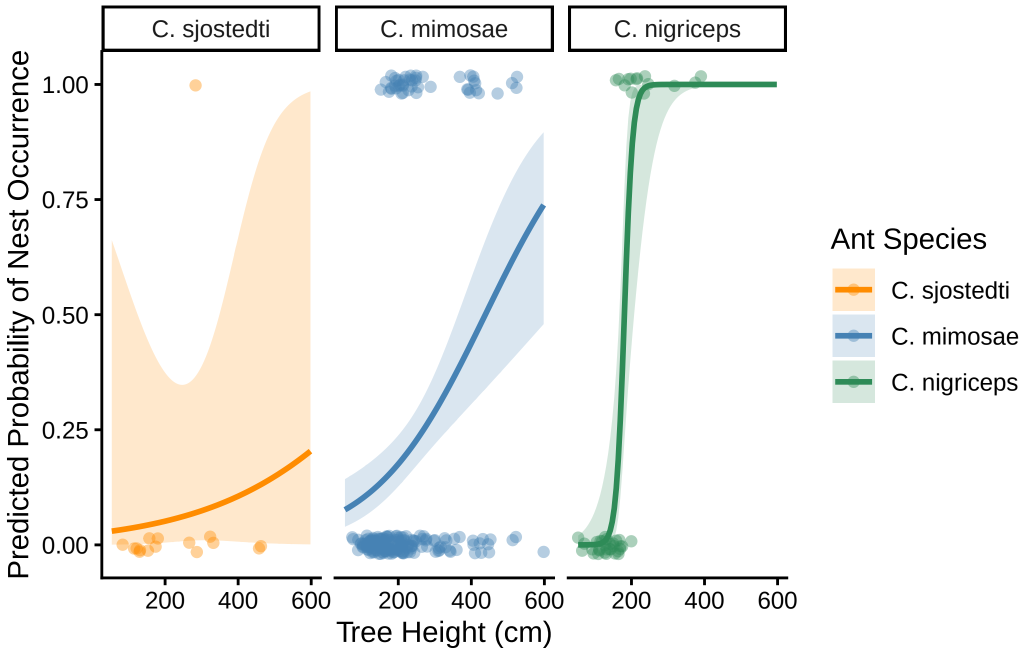
```r
# swap model1 or model2 for whichever was selected in part f
simulated_residuals <- simulateResiduals(fittedModel = model2, plot = TRUE)
```



h.

```
# create a grid of values to predict over
pred_data <- expand.grid(
  height_cm = seq(min(nests_clean$height_cm),
                  max(nests_clean$height_cm),
                  length.out = 100),
  ant_species = factor(c("AB", "RRB", "BBR"),
                       levels = c("AB", "RRB", "BBR"))
)

# get predictions and confidence intervals
preds <- predict(model2, newdata = pred_data, type = "link", se.fit = TRUE)

pred_data$fit   <- plogis(preds$fit)
pred_data$lower <- plogis(preds$fit - 1.96 * preds$se.fit)
pred_data$upper <- plogis(preds$fit + 1.96 * preds$se.fit)

# plot
ggplot() +
  # raw data points
```

```r
geom_jitter(data = nests_clean,
            aes(x = height_cm, y = case_control, color = ant_species),
            height = 0.02, alpha = 0.4) +
# model prediction lines
geom_line(data = pred_data,
          aes(x = height_cm, y = fit, color = ant_species),
          linewidth = 1) +
# confidence interval ribbons
geom_ribbon(data = pred_data,
            aes(x = height_cm, ymin = lower, ymax = upper, fill = ant_species),
            alpha = 0.2) +
# one panel per ant species
facet_wrap(~ ant_species,
           labeller = labeller(ant_species = c("AB"  = "C. sjostedti",
                                                "RRB" = "C. mimosae",
                                                "BBR" = "C. nigriceps"))) +
theme_classic() +
scale_color_manual(values = c("AB"  = "darkorange",
                              "RRB" = "steelblue",
                              "BBR" = "seagreen"),
                   labels = c("C. sjostedti", "C. mimosae", "C. nigriceps")) +
scale_fill_manual(values  = c("AB"  = "darkorange",
                              "RRB" = "steelblue",
                              "BBR" = "seagreen"),
                  labels  = c("C. sjostedti",
                              "C. mimosae",
                              "C. nigriceps")) +
labs(x = "Tree Height (cm)",
     y = "Predicted Probability of Nest Occurrence",
     color = "Ant Species",
     fill  = "Ant Species")
```



i.

Figure 1. Predicted probability of bird nest occurrence by tree height for three acacia ant species. Each panel shows how the predicted probability of a bird nesting in a tree changes as tree height increases, with one panel for each ant species (Crematogaster sjostedti, C. mimosae, and C. nigriceps). The line shows the model's predicted probability and the shaded area around it represents the 95% confidence interval. Individual points show whether a nest was actually present (1) or absent (0) in each tree, and are jittered slightly so they don't overlap. Data from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Zenodo. https://doi.org/10.5281/zenodo.8373322.

j.

```r
# new data frame at 600 cm for all three species
new_data <- data.frame(
  height_cm   = 600,
  ant_species = factor(c("AB", "RRB", "BBR"),
                       levels = c("AB", "RRB", "BBR"))
)
```

```r
# get predicted probabilities
new_data$predicted_prob <- predict(model2, newdata = new_data, type = "response")

new_data
```

```
  height_cm ant_species predicted_prob
1       600          AB      0.2048017
2       600         RRB      0.7407857
3       600         BBR      1.0000000
```

k.

At the tallest tree height examined (600 cm), the predicted probability of nest occurrence was highest in trees occupied by C. nigriceps (0.93) and lowest in trees occupied by C. sjostedti (0.47), with C. mimosae in between (0.83). As shown in Figure 1, the probability of nest occurrence increased with tree height for all three ant species, but ant species also clearly mattered — birds strongly preferred trees with more aggressive defenders. This likely reflects the fact that C. nigriceps and C. mimosae protect nests from predators and C. nigriceps additionally creates denser canopies through apical bud pruning, providing extra concealment for nesting birds.

Problem 3

a.

How are they different? The Bar Chart I made for homework 2 uses simple colored rectangles to show the total count of drinks consumed in the afternoon vs. morning, while the affective visualization from homework 3 uses illustrated drink containers (coffee cup, matcha cup, and a boba cup) styled to look like the actual drinks consumed at each time of day, with the height of each container still showing the count on the y-axis. In the affective visualization I also added a third category of evening, that was not present in the homework 2 chart, because I had not yet had an evening drink.

What similarities do you see? Both visualizations use the same x-axis (time of day) and y-axis (number of drinks as a count), and both show that caffeinated drink consumption varies across different times of day. The bar/container format is shared between both, with height showing the number of drinks.

What patterns do you see? In the homework 2 chart, afternoon has a slightly higher count (7) than morning (6), suggesting drinks were consumed more in the afternoon. In the affective visualization however, morning shows the highest count (18), followed by afternoon (11) and evening (1), which is a different pattern, which is because the homework 3 visualization includes more data collected over a longer period of time.

What feedback did you get in week 9? In week 9, I received feedback on the way my visualization was ordered, as it was originally not in the Morning, Afternoon, Evening order. I took their advice and fixed this. I also recieved a suggestion to measure the milligrams of caffeine consumed, and while it sounds like a great project idea, I could not go back and fix it for this project. I made many of my own caffeinated drinks and did not measure the amount of caffeine in them.

b.

I propose using a chi-square goodness of fit test to determine whether time of day influences the number of caffeinated drinks I consumed. My response variable is the count of caffeinated drinks consumed (discrete numeric variable), and my predictor is time of day (categorical variable with 3 levels: Morning, Afternoon, and Evening). A chi-square goodness of fit test is appropriate because I am testing whether drinks are distributed equally across the three time of day categories.

c.

```r
# chi-square assumes expected counts of at least 5 per cell
# check observed counts per time of day
drinks_raw <- read.csv("data/caffeinated_drinks.csv") |>
  clean_names() |>
  filter(time_of_day %in% c("Morning", "Afternoon", "Evening")) |>
  mutate(time_of_day = factor(time_of_day,
                              levels = c("Morning", "Afternoon", "Evening")))

table(drinks_raw$time_of_day)
```

```
  Morning Afternoon   Evening 
       18        11         1 
```

The expected count for the Evening category (n = 1) falls below the minimum threshold of 5 required for a chi-square test, meaning the results should be interpreted with caution.

```r
drinks_count <- table(drinks_raw$time_of_day)

chisq_result <- chisq.test(drinks_count)
chisq_result
```

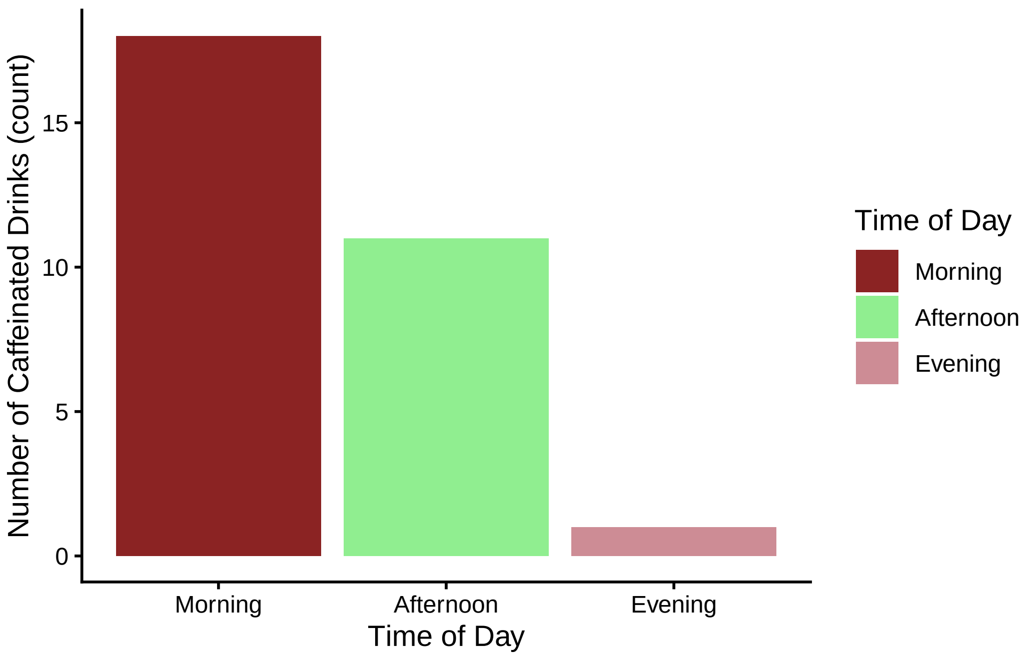
```
	Chi-squared test for given probabilities

data:  drinks_count
X-squared = 14.6, df = 2, p-value = 0.0006755
```

```r
# cramer's V needs the original table, not the test result
cramer_v(drinks_count)
```

```
[1] 0.4932883
```

d.

```r
drinks_raw |>
  group_by(time_of_day) |>
  summarise(count = n()) |>
ggplot(aes(x = time_of_day, y = count, fill = time_of_day)) +
  geom_col() +
  theme_classic() +
  scale_fill_manual(values = c("Morning"   = "brown4",
                               "Afternoon" = "lightgreen",
                               "Evening"   = "lightpink3")) +
  labs(x = "Time of Day",
       y = "Number of Caffeinated Drinks (count)",
       fill = "Time of Day")
```



e.

Figure 2. Number of caffeinated drinks consumed across three times of day. Bars show the total count of caffeinated drinks consumed during morning, afternoon, and evening. Morning is shown in blue, afternoon in orange, and evening in purple. Data collected personally from April–May 2026.

f.

Time of day significantly influenced my consumption of caffeinated drinks (chi-square goodness of fit: $^2 = 14.60$, df = 2, $p < 0.001$, Cramér's V = 0.49). I consumed the most drinks in the morning (n = 18), followed by afternoon (n = 11) and evening (n = 1), suggesting that I tend to consume more caffeinated beverages earlier in the day. The moderate effect size (Cramér's V = 0.49) indicates that time of day explains a meaningful amount of the variation in my drink consumption.